

# From whales to minnows: The impact of crypto-reward fairness on user engagement in social media

Woojin Yang<sup>a</sup>, Yeongin Kim<sup>b</sup>, Tae Hun Kim<sup>c</sup>, Chul Ho Lee<sup>d,\*</sup>, Yasin Ceran<sup>e</sup>

<sup>a</sup> College of Business, Korea Advanced Institute of Science and Technology, 291 Daehakro, Yuseong-Gu, Daejeon 34141, Republic of Korea

<sup>b</sup> Wake Forest School of Business, Wake Forest University, 1834 Wake Forest Rd., Winston-Salem, NC 27106, United States of America

<sup>c</sup> College of Business Administration, Incheon National University, 119 Academy-ro, Yeonsu-gu, Incheon 22012, Republic of Korea

<sup>d</sup> College of Business, Korea Advanced Institute of Science and Technology, 291 Daehakro, Yuseong-Gu, Daejeon 34141, Republic of Korea

<sup>e</sup> School of Information Systems and Technology, Lucas School of Business, San Jose State University, One Washington Square, San Jose, CA 9519, United States of America

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## ABSTRACT

In an era where user-generated content drives social media growth, effectively incentivizing contributions remains a challenge. This study explores the empirical impact of a crypto-integrated platform, Steemit, focusing on a system transition designed to enhance fairness in reward distribution. We assess how this shift affects user engagement, specifically through the volume of posts. Our findings indicate that a fairer crypto-reward distribution boosts user-generated posts, though the increase is less pronounced for users with higher capital or reputation. Further analysis reveals the complex dynamics of cryptocurrency rewards and their role in fostering individual contributions and platform growth, while offering financial incentives. The effects of fair distribution are consistent across diverse user groups, highlighting novel incentivization strategies in social media and the transformative potential of integrating cryptocurrencies into reward systems.

## 1. Introduction

The integration of cryptocurrencies is emerging as an innovative solution to promote user participation in social media. This is seen in platforms such as Steemit, Medium, Scorum, and SocialX, which have incorporated cryptocurrencies as rewards for user content creation and sharing. This novel approach aims to enhance user participation by making a tangible connection between engagement and financial gain. The essence of this incentive lies in a unique relationship between user engagement, platform growth, and the financial value of the platform-specific cryptocurrency. As users engage and consume more content, the platform expands, which positively influences the cryptocurrency's value [1]. This model encourages users to actively contribute to the platform because their success could increase their cryptocurrency holdings' value. As such, cryptocurrency incentives have the potential to play a pivotal role in promoting user engagement and participation within social media platforms [1].

Expanding upon the idea of cryptocurrency incentives to stimulate user engagement, the role of fairness in the distributive rules cannot be

overstated. These rules, which determine the allocation of rewards among content creators, significantly influence user engagement. A user's motivation to participate actively hinges on their belief in the fairness of the reward distribution [2]. By implementing distributive rules that are perceived as fair, a platform can foster an environment where contributions are proportionately rewarded, creating a more engaged community. Therefore, to truly realize the potential of cryptocurrency incentives, it is crucial to examine to the aspect of fairness in the creation and application of distributive reward rules.

In this paper, we examine the effects of fair reward distribution in crypto-based incentive systems on user engagement within social media platforms. Our framework defines fairness as the equitable distribution of rewards, aligning with distributive justice principles [3]. Although cryptocurrency rewards are increasingly used to enhance user engagement, research on how fairness affects user behavior remains limited [4,5]. We utilize a case study from Steemit, which revised its reward mechanism to favor less dominant users, thereby promoting fairness. This study investigates how such changes impact user participation in cryptocurrency-integrated platforms. In specific, our research addresses

\* Corresponding author.

E-mail addresses: [dnwls2669@kaist.ac.kr](mailto:dnwls2669@kaist.ac.kr) (W. Yang), [kimy@wfu.edu](mailto:kimy@wfu.edu) (Y. Kim), [taehunkim@inu.ac.kr](mailto:taehunkim@inu.ac.kr) (T.H. Kim), [irontigerlee@kaist.ac.kr](mailto:irontigerlee@kaist.ac.kr) (C.H. Lee), [yasin.ceran@jsu.edu](mailto:yasin.ceran@jsu.edu) (Y. Ceran).

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two main questions: “Does the shift in distributive rule increase the volume of content created by users?” and “Which user characteristics moderate this effect?” Given the stagnation in growth many platforms face, understanding the impact of distributive rules is vital for their success. We specifically explore how capital and reputational resources—indicative of a user’s financial power and social influence, respectively—affect engagement and content quality, closely relating to organizational justice and behavior within it [6,7].

We utilize a unique dataset from Steemit, a leading crypto-integrated social media platform, focusing on the introduction of Hardfork19 (HF19) on June 2017, as a significant event. HF19 changed Steemit’s reward function from a quadratic to a linear model, equalizing the influence of each account on reward distribution and aiming to reduce the dominance of large capital holders, ‘Whale users.’ This shift, termed *Equality*, marks a notable change in Steemit’s approach to rewards, offering a distinct setting for studying the effects of fair crypto-based incentives on user engagement [8].

Our analysis uses a panel dataset of 2996 users from ten weeks before and after HF19’s implementation, tracking activities like posting, following, and transactions. By leveraging econometric models, we evaluate how this new reward policy influences user post volume. We further explore how user characteristics, particularly capital and reputational resources, might moderate these effects. This moderation analysis helps understand different user responses to the revised reward system, providing deeper insights into fairness and user engagement in crypto-integrated platforms.

Our findings demonstrate that the volume of user posts significantly increased following the implementation of HF19, which implies that this shift towards equitable distribution rules amplifies user contributions to the platform. However, we observe that users’ capital and reputational resources serve as significant moderators of this effect. Specifically, users with fewer capital resources are found to be more motivated, creating a greater number and lengthier posts. Exploring the intricacies of user engagement specific to crypto-reward social media platforms, we show that after the transition, users were not just more inclined to prefer a cryptocurrency that is non-tradable and used only within the platform. This indicates a long-term investment in the platform over short-term monetary profits. Moreover, the quality of content generated post-transition significantly improved after HF19. Interestingly, these trends are not confined to any specific demographic; users with ample financial or reputational resources also exhibited similar behaviors influenced by HF19.

In summary, our research contributes to distinct yet interconnected streams of literature — social media, token economy, and distributive justice. Previous studies on social media have mostly focused on the effects of traditional financial rewards like cash or coupons [9,10,11,12,13]. Recent investigations have considered the broader impact of such rewards on user-generated content [10,12,13], but the influence of reward distribution has not been fully explored. Our work differs by examining a crypto-integrated incentive system where the value of rewards is closely tied to content creation and engagement, emphasizing the need for fair reward distribution for sustainable social media success. We provide empirical evidence that fairness in reward distribution enhances content creation and broaden the distributive justice literature by applying these principles to online communities. The issue of power centralization among influential ‘whale users’ remains unresolved. Our findings contribute insights for developing equitable crypto-based incentives in digital platforms, aiding stakeholders in the creation of data-driven decision-making tools. These tools help implement incentive systems that enhance engagement and ensure long-term platform sustainability by integrating fairness and justice into algorithms, thus positively shaping user behavior and platform dynamics.

## 2. Research background

### 2.1. Financial incentive in social media

Financial incentives are a key motivator for user engagement in social media platforms, successfully implemented by YouTube, Twitch, Facebook, and Instagram through ad revenue sharing, direct payments, affiliate programs, and sponsored content partnerships [14]. These mechanisms hinge on the idea that financial gain boosts user engagement, drives content creation, and nurtures vibrant communities [15]. From a user perspective, financial rewards encourage more active participation and content curation, thereby fostering a competitive and engaging community environment [16,9,10,17,18,19,20]. The lure of financial rewards deepens and diversifies the content, enhancing the platform’s appeal [21]. Understanding these dynamics is vital for comprehending how social media evolves [22].

However, the shift to cryptocurrency rewards marks a significant change, linking individual efforts, community growth, and financial outcomes. Crypto incentives highlight the dual role of users as participants and stakeholders [23]. Our study expands on this by examining the effects of HF19, which altered distributive rules towards more equitable reward distribution, assessing its impact on user content creation and platform sustainability.

Building on this understanding, our study goes further than observing active participation based on user characteristics. We leverage HF19 as an external shock that changed the distributive rules, a critical element in blockchain incentive systems. Our goal is to examine the impact of the more equitable reward distribution rule on the content-creation of users of varying statuses. Additionally, we aim to assess how these changes influence the overall sustainability of the platform.

### 2.2. Cryptocurrency rewards in social media

With an advance of distributed ledger technology, *blockchain*, cryptocurrency rewards have been a recent and transformative development in the field of social media [1]. Emerging as a solution to counteract declining user engagement rates, cryptocurrencies offer a novel method to incentivize content creation and sharing on social media platforms. This system essentially monetizes user participation, thereby linking active contributions to potential financial gain. It has recently been implemented by various platforms such as Steemit, Medium, Scorum, and SocialX, fundamentally altering the dynamics of user interaction. The mechanism behind this system is fairly straightforward: users are rewarded with platform-specific cryptocurrencies for generating and sharing content.

Within our research context, Steemit, three types of cryptocurrencies are utilized. The foundational currency is STEEM, an integral part of the Steemit platform, stemming from the platform’s incentive system (i.e., blockchain). Steem Power (SP) is the platform’s native currency, not directly tradable on exchanges, but used to amplify users’ influence when voting and posting content. The Steem Dollar (SBD) is another reward currency traded on exchanges, which maintains a stable face value pegged to the U.S. dollar. Within the platform, STEEM, SP, and SBD are exchangeable with each other. For content creation (e.g., posting on the user’s blog page), users are rewarded with SP and SBD. The relationships among the three digital currencies in Steemit are illustrated in Fig. 1. (See Table 1.)

The interconnectedness of user contributions, platform growth, and crypto value is central to the dynamics of crypto-integrated platforms. Here, user contributions are essential for growth. For example, high-quality posts by Steemit users not only attract a broader audience but also encourage more user interactions and engagement. As activity increases, the demand for platform currencies grows, raising their value and directly benefiting content creators. This cycle of contributions, growth, and financial rewards motivates users further, linking individual achievements to the platform’s success. For instance, earning SP

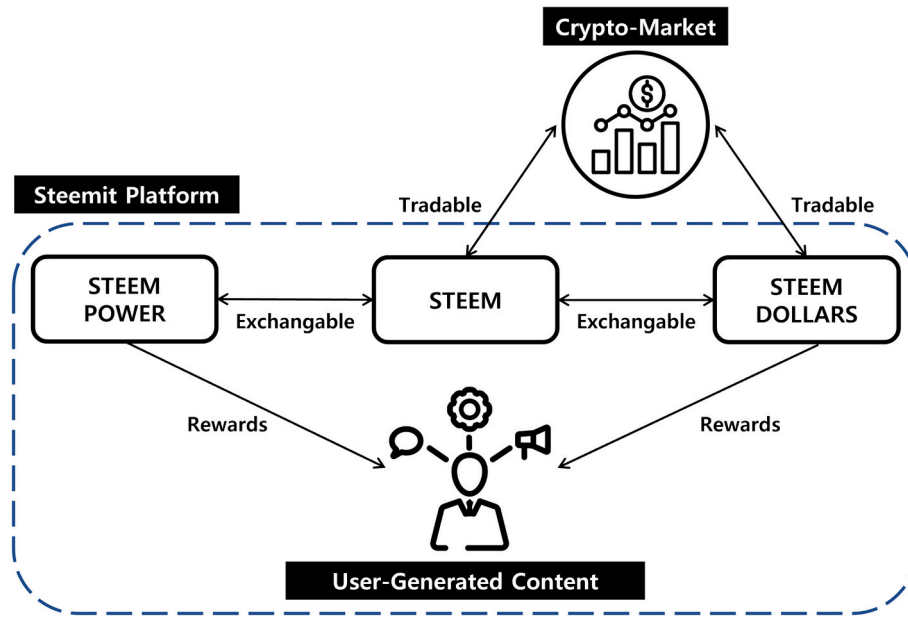


Fig. 1. Relation of three digital currencies in steemit.

**Table 1**  
Variables and summary statistics.

| Variable            | Description                                                                               | N      | Mean    | Std.Dev. | Min    | Max     |
|---------------------|-------------------------------------------------------------------------------------------|--------|---------|----------|--------|---------|
| <i>Post</i>         | The number of posts generated by user $i$ in time $t$                                     | 62,916 | 2.669   | 5.717    | 0      | 167     |
| <i>Log(Length)</i>  | The Log total number of characters for posts generated by user $i$ in time $t$            | 29,360 | 3.960   | 4.362    | 0      | 13.582  |
| <i>Treat</i>        | = 1 if $t$ year-week is after 2017-25th week initialized HF19; 0 otherwise.               | 62,916 | 0.524   | 0.499    | 0      | 1       |
| <i>Tenure</i>       | Days since user $i$ creates account in time $t$                                           | 62,916 | 246.301 | 109.767  | 7      | 525     |
| <i>Following</i>    | The number of following user $i$ applies in time $t$                                      | 62,916 | 0.448   | 12.522   | 0      | 1344    |
| <i>Followed</i>     | The number of new followers of user $i$ made in time $t$                                  | 62,916 | 1.983   | 20.595   | 0      | 930     |
| <i>Delegation</i>   | The number of delegating user $i$ to other users in time $t$                              | 62,916 | 0.047   | 0.928    | 0      | 119     |
| <i>Conversion</i>   | The number of applications of conversion SBD to STEEM by user $i$ in time $t$             | 62,916 | 0.094   | 0.717    | 0      | 52      |
| <i>Withdrawal</i>   | The number of withdrawals from Steem Wallet to external exchanges by user $i$ in time $t$ | 62,916 | 0.378   | 1.475    | 0      | 74      |
| <i>Deposit</i>      | The number of deposits from external exchanges to Steem Wallet by user $i$ in time $t$    | 62,916 | 0.145   | 1.135    | 0      | 87      |
| <i>Log(SP)</i>      | The Log amounts of Steem Power that user 62,916 $i$ holds in time $t$                     | 62,916 | 22.08   | 3.507    | 0      | 30.051  |
| <i>Log(Rep)</i>     | The Log number of Reputation score of 62,916 user $i$ in time $t$                         | 62,916 | 27.808  | 4.135    | 0      | 34.190  |
| <i>STMPrice</i>     | STEEM token closed price in time $t$                                                      | 62,916 | 1.222   | 0.574    | 0.216  | 2.454   |
| <i>Log(STMVol)</i>  | Log number of STEEM token volume traded within time $t$                                   | 62,916 | 16.712  | 0.811    | 15.173 | 18.755  |
| <i>STMVar</i>       | Variation (difference between the highest and the lowest) of STEEM token within time $t$  | 62,916 | 0.559   | 0.325    | 0.066  | 1.520   |
| <i>STMPriceRate</i> | The increase rate of STEEM token's closed price in time $t$ compared with time $t - 1$    | 62,916 | 0.852   | 2.171    | -0.412 | 8.328   |
| <i>STMVolRate</i>   | The increase rate of STEEM token's volume traded in time $t$ compared with time $t - 1$   | 62,916 | 5.524   | 21.705   | -0.730 | 102.345 |

provides not just financial benefits but also greater voting power and earning potential on the platform. This symbiotic relationship between users and the platform enhances user engagement and sustains growth, showcasing an innovative model to boost participation.

### 2.3. Rewards distribution and distributive justice

Reward distribution and distributive justice are cornerstone concepts that play crucial roles in shaping behaviors in numerous settings, from traditional organizations to online communities [24]. Distributive justice involves the equitable and just allocation of outcomes and rewards at an organizational level [3]. These principles gain particular importance in contexts where the participation and contributions of individuals are vital [25].

In traditional organizations, the principles of distributive justice often inform compensation structures, influencing how employees perceive their rewards relative to their efforts [26]. It is a delicate balance that requires a fair distribution of rewards to motivate employees and encourage productivity. Moving this framework to the digital realm of social media, the same principle applies [27]. However, with the

advent of cryptocurrency rewards, the dynamics of reward distribution have changed significantly [5]. The introduction of blockchain technology has enabled a more transparent, automated, and rule-based distribution of rewards, potentially increasing the perceived fairness of reward distribution [5]. The notion of distributive justice in this context becomes complex, as it is not merely about dividing a fixed pie but rather about how the distribution rules can influence the growth of platform and the value of rewards themselves [1].

The intersection of reward distribution and distributive justice on social media platforms, especially those employing cryptocurrency incentives, heralds uncharted areas of research. This interplay invites a thorough investigation into the effects of fairness in reward distribution on user behavior, the dynamics of user interaction, and the overall growth trajectory of the platform. Our research context offers a unique empirical setting that underscores a notable case of a reward system transition within a crypto-integrated social media platform. Within the framework of various justice principles and distribution rules, we consider HF19 as embodying the *contributions rule*, which focuses on the equality of outcome/input ratios or equality relative to individual contributions [28]. The linear reward function introduced by HF19 on the

Steem platform well exemplifies the perspective, ensuring that rewards are proportionally distributed based on users' investments (Steem Power), thereby achieving distributive justice.

### 3. Hypothesis development

When examining the dynamics of cryptocurrency rewards on social media platforms, it is crucial to consider the influence of fairness in distribution on user behavior, especially concerning content creation. We argue that fair distribution significantly influences the quantity of user-generated content. The logic behind this hypothesis lies in both the incentives provided by a fair rewards system and the psychological impact it entails, as suggested by the literature.

Fairness Theory posits that perceived fairness leads to a range of positive outcomes, including increased satisfaction, motivation, and positive affect [2]. This theory hinges on the notion that individuals assess fairness based on the outcomes they receive, the processes leading to these outcomes, and how these processes are communicated. Translated into the context of a social media platform, when users perceive the distribution of cryptocurrency rewards as fair, they feel valued and respected. This enhanced perception of self-worth motivates them to contribute more actively to the platform, such as by creating and sharing content.

Self-determination theory suggests that intrinsic motivation is enhanced when three core psychological needs—autonomy, competence, and relatedness—are met [29,30]. On blockchain-based social media platforms, fairness in crypto-reward distribution is crucial. A fair and transparent allocation mechanism increases users' autonomy, as they view the reward system as predictable and controllable. It also supports competence by ensuring that contributions are recognized and rewarded. Additionally, since rewards are often based on community engagement, they naturally promote relatedness. Thus, fair distribution of crypto rewards taps into these intrinsic motivators, encouraging wider active participation. Guided by this framework, we propose:

**Hypothesis 1.** Fair distribution of crypto-rewards for content creation is positively associated with the amount of content created by a user.

We further delve deeper into the potential moderation effects of user resources on the impact of fair crypto-rewards distribution. We focus on two types of resources: capital resources and reputation resources. Capital resources refer to the user's financial power in terms of cryptocurrency holdings. On the other hand, reputation resources signify the user's influence earned by generating quality content and engaging with the platform. We hypothesize that the presence of these resources might influence the relationship between fair reward distribution and the quantity of user-generated content.

In the Steemit context, users with substantial capital, often called Whale users, have significant influence due to their large cryptocurrency holdings. A move towards a more equitable reward distribution could decrease their rewards, potentially affecting their extrinsic motivation and altering their contribution behavior. Expectancy Theory suggests individuals act when they anticipate that their actions will lead to desired outcomes [31]. Whale users, likely motivated extrinsically by significant investments, may find a fairer reward distribution less motivating. Additionally, they could see this change as undermining their dominance and control, negatively impacting their intrinsic motivation by affecting their sense of autonomy and competence [29]. According to Fairness Theory, they might also perceive the new reward system as unjust if it reduces their rewards disproportionately to their inputs, leading to lower satisfaction and reduced contribution [2]. Therefore, we propose:

**Hypothesis 2a.** The positive effect of fair crypto-rewards distribution on the amount of user-generated content decreases when it is applied to users with more capital resources.

Similarly, users with a lot of reputational resources have shown that

they are dedicated by taking an active part in the platform. These users have built up a lot of trust and sway within the online community. However, when rewards are distributed more evenly, their reputation and contributions might not bring in the same rewards as before. This could be seen as a risk to their current standing on the platform. According to the Self-Determination Theory, this perceived risk could lower their sense of capability, which in turn could lessen their internal motivation [29].

Furthermore, in light of Fairness Theory, which emphasizes individuals' perceptions of equitable treatment, these high-reputation users might view the change in reward distribution as unfair. This perceived unfairness, especially if they feel their contributions are no longer being rewarded as they used to be, can lead to dissatisfaction and a decrease in extrinsic motivation. The theory suggests that when individuals perceive an imbalance between their input (e.g., effort, reputation) and output (e.g., financial rewards), their motivation to engage in the contributing behavior can decrease. Thus, the adjustment in reward distribution, while aiming for overall fairness, might inadvertently impact the extrinsic motivation of users who previously benefited from a more skewed reward system. Therefore, we propose that while equitable reward distribution intends to foster fairness, it may have unintended consequences on the motivation of users with high reputational resources, particularly in terms of their extrinsic motivation. This consideration is crucial in understanding the complex dynamics of user engagement in response to changes in reward distribution policies on social media platforms.

**Hypothesis 2b.** The positive effect of fair crypto-rewards distribution on the amount of user-generated content decreases when it applies to users with more reputation resources.

### 4. Research setting, data, and variables

Our data comes from Steemit, a leading crypto-integrated social media platform. Steemit rewards users with cryptocurrency for their content creation activities. Central to this platform is its native cryptocurrency, STEEM, and its two derivative currencies, Steem Power (SP) and Steem Dollars (SBD). The platform's unique incentive structure rewards users with SP and SBD for creating posts and for upvoting others' content. This vested, long-term investment in the platform confers users with influence over the distribution of rewards, creating a compelling dynamic between individual contributions and potential financial gains.

Steemit's initial reward distribution mechanism faced criticism for its *quadratic* rewards curve. This structure disproportionately favored users with larger amounts of Steem Power (SP), incentivizing investment in SP, but in turn, created a *rich-get-richer* environment. This system established barriers for new users, referred to as *Minnows*, even those producing high-quality content. In other words, this ecosystem, heavily skewed in favor of a few influential users, called *Whales*, led to a lack of fairness, prompting the platform developers to reassess their reward distribution system [32]. In response, they introduced Hardfork19 (HF19) on June 6th, 2017. HF19 is an adjustment to the influence of users on reward amounts and is designed to dilute the dominance of whale users. As reflected in its code name *Equality*, this transition aimed to bolster the incentive for a broader user population, fostering a more balanced and inclusive community.<sup>1, 2 and 3</sup> We utilize the introduction of HF19 as a shock to empirically examine the fairness in the distribution of crypto rewards on users' content creation in social media.

<sup>1</sup> HF19 introduced several updates, including notable modifications to the voting impact and curation rewards [33]. However, according to the developers' remarks, these updates were considered minor. A detailed discussion of these aspects is provided in Section 6.4.



#### 4.1. Sample construction

We examine users who were active between the 15th and 25th weeks of 2017. More specifically, we consider users who published a post, a comment, or cast a vote on other users' content at least once during this observation period. Detailed activity data for each user was obtained through the website's APIs (accessed in July 2019). All data linkages are established using two unique identifiers: an activity identification number, generated each time a user engages in activities like posting, receiving rewards, or transacting cryptocurrency. We consolidate this granular data into a panel dataset organized by user on a weekly basis. Our final sample consists of 2996 unique users over 21 weeks, ten weeks before and after the implementation of HF19, resulting in a balanced panel of 62,916 user-week observations.

#### 4.2. Variables

To evaluate the impact of fair reward distribution, we employ two key measures in alignment with recent studies [9,34]. The first measure, denoted as *Post*, quantifies the number of posts created by a user in a given week. We also measure *Length*, which captures the total length of contents the user generated. Notably, to mitigate the impact of *Length*'s skewed distribution in our sample, we transform this variable using the formula  $\log(1 + n)$ .

In our analysis, the primary independent variable, *Treat*, indicates the implementation of HF19. It is a binary variable, set to 1 from the week HF19 was introduced. We include two moderators to explore how different factors might influence the impact of fair reward distribution on content creation. The first, *SP*, measures a user's capital resources by quantifying their average weekly SP holdings, reflecting their financial influence in voting and overall capital resources. The second, *Rep*, represents the user's reputation within the Steem community, calculated by Steemit to reflect the perceived quality of their contributions and trust from other users. High *Rep* scores indicate high-quality content and greater community trust. We log-transform these moderators to address their skewed distributions.

To account for other variations, we include several user-level control variables: days since account creation (*Tenure*); number of users followed (*Following*); number of followers (*Followed*); users delegated for voting (*Delegation*); frequency of SBD to STEEM conversions (*Conversion*); number of withdrawals (*Withdrawal*); and number of deposits (*Deposit*). We also control for market-level variables affecting user engagement: STEEM token's closing price (*STMPrice*); log-transformed STEEM trading volume (*STMVol*); weekly STEEM price variation (*STMVar*); and weekly rates of STEEM price and volume increases (*STMPriceRate*, *STMVolRate*). Preliminary analysis without a model (Fig. 2) shows a significant increase in content volume post-HF19, with average posts per user rising from 2.5 to 5.4 and total length from 1200 to 2100 characters. This suggests that fairer distribution may boost content creation. Yet, other factors like crypto market trends could also influence results. We follow up with a detailed empirical analysis to discern the effects of fair distribution more clearly.

### 5. Main analysis

#### 5.1. Main treatment effect

We first examine the main treatment effect of fair rewards distribution on the quantity of contents individual user *i* created during week *t*, by specifying the following regression equation:

$$y_{it} = \beta_0 + \beta_1 \text{Treat}_t + \gamma X_{it} + \delta Z_t + \mu_i + \epsilon_{it} \quad (1)$$

Our dependent variable,  $y_{it}$ , represents  $\text{Post}_{it}$ , the number of posts created by user *i* in week *t*. As a robustness check, we also consider  $\text{Length}_{it}$  that captures the total length of text contents of user *i* in week *t* as an alternative dependent variable.  $\text{Treat}_t$  is a dummy variable

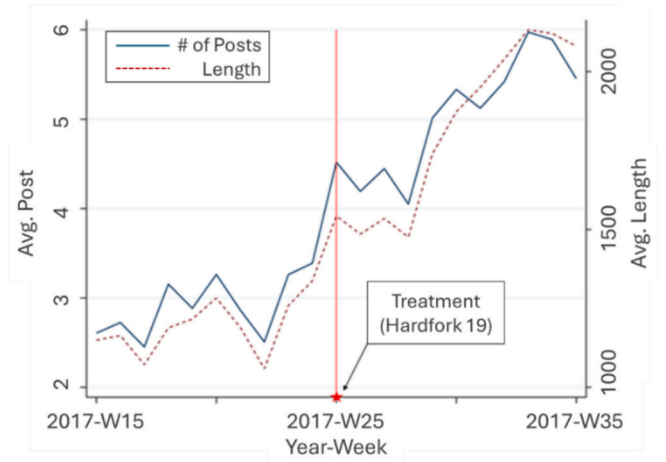


Fig. 2. Trends in user's average post volume before and after Hardfork19. Note: The solid blue line indicates the average number of posts over time; The dashed red line indicates the average post length over time. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

indicating if the observations occurred after HF19 was introduced.  $X_{it}$  is a vector of user-level control variables, and  $Z_t$  is a vector of control variables related to the cryptocurrency market. We also consider a user-level fixed effect  $\mu_i$ .

We demonstrate the main treatment effect of HF19 on the number of posts created by individual users by regression models in Eq. (1). As our main dependent variable,  $\text{Post}_{it}$ , is a count variable, we apply the Poisson and Negative Binomial (NB) models and report the results in columns (1)–(4) of Table 2. We also report the regression results with the alternative dependent variable,  $\text{Length}_{it}$ , in columns (4)–(6) of Table 2. Our results reveal a significant increase in the number of posts created by an individual user after the introduction of HF19, as reported in column (2) of Table 2 ( $\beta = 0.368, p < 0.01$ ). The result remains consistent even after controlling for a variety of control variables, as shown in columns (2)–(4) of Table 2 ( $\beta = 0.630, p < 0.01; \beta = 0.485, p < 0.01$ ). These results imply that the implementation of HF19 is associated with a 1.878 increase in the number of posts by a user. In addition, we run the model with an alternative dependent variable, *Length*, and find similar observations that after the introduction of HF19, the quantity of user participation increases. The homogeneous trend across the models gives evidence that supports our main hypothesis, H1.

#### 5.2. Heterogeneous treatment effect

We next examine the heterogeneous treatment effect of focal interest - the fair distribution of crypto-rewards, as specified in the following equation:

$$y_{it} = \beta_0 + \beta_1 \text{Treat}_t + \beta_2 \text{UserStatus}_{it} + \beta_3 \text{Treat}_t \times \text{UserStatus}_{it} + \gamma X_{it} + \delta Z_t + \mu_i + \epsilon_{it}, \quad (2)$$

where  $\text{UserStatus}_{it}$  is either  $\text{SP}_{it}$  or  $\text{Rep}_{it}$  where  $\text{SP}_{it}$  captures the amount of SP that individual user *i* holds and  $\text{Rep}_{it}$  is the average reputation score of user *i* over week *t*.

We first focus on the moderating role of users' financial resources, as denoted by the  $\text{SP}_{it}$ . The results, presented in column (2) of Table 3, suggest that users' financial resources significantly attenuate the impact of fair reward distribution. The coefficient estimate of  $-0.072$  for  $\text{Treat} \times \text{SP}$  is both negative and significant, implying a marked decrease in the number of posts, with an increment in SP. The application of an alternative regression model, Negative Binomial, yields a qualitatively consistent result, as shown in column (3) of Table 3. This indicates that users with more capital resources are less responsive to a fair

**Table 2**  
Effects of HF19 on the quantity of user posts.

| Variable             | Post                |                      |                      | Length              |                      |
|----------------------|---------------------|----------------------|----------------------|---------------------|----------------------|
|                      | (1) Poisson         | (2) Poisson          | (3) NB               | (4)                 | (5)                  |
| <i>Treat</i>         | 0.368***<br>(0.025) | 0.630***<br>(0.031)  | 0.485***<br>(0.017)  | 1.159***<br>(0.052) | 1.759***<br>(0.069)  |
| <i>Log(SP)</i>       |                     | 0.041***<br>(0.011)  | 0.079***<br>(0.004)  |                     | 0.098***<br>(0.009)  |
| <i>Log(Rep)</i>      |                     | 0.041**<br>(0.018)   | 0.107***<br>(0.005)  |                     | 0.054***<br>(0.016)  |
| <i>Following</i>     |                     | 0.002***<br>(0.000)  | 0.002***<br>(0.000)  |                     | 0.005***<br>(0.001)  |
| <i>Followed</i>      |                     | −0.001<br>(0.001)    | −0.000**<br>(0.000)  |                     | 0.003**<br>(0.002)   |
| <i>Delegation</i>    |                     | −0.010<br>(0.007)    | −0.000<br>(0.004)    |                     | 0.003<br>(0.015)     |
| <i>Conversion</i>    |                     | 0.044**<br>(0.020)   | 0.054**<br>(0.003)   |                     | 0.426***<br>(0.094)  |
| <i>Withdrawal</i>    |                     | 0.011*<br>(0.007)    | 0.022***<br>(0.002)  |                     | 0.153***<br>(0.033)  |
| <i>Deposit</i>       |                     | 0.012*<br>(0.006)    | 0.015***<br>(0.002)  |                     | 0.189***<br>(0.069)  |
| <i>Tenure</i>        |                     | −0.004***<br>(0.000) | −0.001***<br>(0.000) |                     | −0.013***<br>(0.001) |
| <i>STMPPrice</i>     |                     | 0.239***<br>(0.020)  | 0.245***<br>(0.016)  |                     | 0.886***<br>(0.046)  |
| <i>Log(STMVol)</i>   |                     | 0.231***<br>(0.014)  | 0.285***<br>(0.014)  |                     | 0.408***<br>(0.034)  |
| <i>STMVar</i>        |                     | −0.006<br>(0.029)    | −0.111***<br>(0.027) |                     | 0.092<br>(0.072)     |
| <i>STMPPriceRate</i> |                     | −0.012***<br>(0.004) | −0.009**<br>(0.004)  |                     | −0.017**<br>(0.008)  |
| <i>STMVolRate</i>    |                     | 0.000<br>(0.000)     | −0.000<br>(0.000)    |                     | −0.001<br>(0.001)    |
| Fixed Effects        | Yes                 | Yes                  | Yes                  | Yes                 | Yes                  |
| Log-likelihood       | −114,810            | −108,065             | −85,855              | −                   | −                    |
| R-squared            | −                   | −                    | −                    | 0.018               | 0.104                |
| Observations         | 62,916              | 62,916               | 62,916               | 29,360              | 29,360               |

Notes. Standard errors in parentheses. \*\*\*  $p < 0.01$ , \*\*  $p < 0.05$ , \*  $p < 0.1$ .

**Table 3**  
Heterogeneous effects of HF19 on the quantity of user posts.

| Moderators            | SP                   |                      | Rep                  |                      |
|-----------------------|----------------------|----------------------|----------------------|----------------------|
|                       | (1) Poisson          | (2) NB               | (3) Poisson          | (4) NB               |
| <i>Treat</i>          | 2.265***<br>(0.348)  | 2.666***<br>(0.097)  | 1.977***<br>(0.362)  | 2.152***<br>(0.129)  |
| <i>Log(SP)</i>        | 0.075***<br>(0.020)  | 0.129***<br>(0.004)  | 0.038***<br>(0.011)  | 0.074***<br>(0.004)  |
| <i>Log(Rep)</i>       | 0.039**<br>(0.016)   | 0.180***<br>(0.006)  | 0.070***<br>(0.020)  | 0.215***<br>(0.006)  |
| <i>Treat×Log(SP)</i>  | −0.072***<br>(0.015) | −0.095***<br>(0.004) |                      |                      |
| <i>Treat×Log(Rep)</i> |                      |                      | −0.045***<br>(0.012) | −0.056***<br>(0.004) |
| Controls              | Yes                  | Yes                  | Yes                  | Yes                  |
| Fixed Effects         | Yes                  | Yes                  | Yes                  | Yes                  |
| Log-likelihood        | −107,468             | −85,594              | −107,758             | −5767                |
| Observations          | 62,916               | 62,916               | 62,916               | 62,916               |

Notes. Standard errors in parentheses. \*\*\*  $p < 0.01$ , \*\*  $p < 0.05$ , \*  $p < 0.1$ .

distribution of rewards, potentially due to their already high capacity to earn rewards. Next, we examine the moderating effect of reputation resources on the relationship between fair reward distribution and users' content production and present the results in column (4) of Table 3. We find that the positive effect of HF19 is significantly reduced when the user has a higher reputation score. The negative and significant coefficient estimate of  $-0.045$  for  $Treat \times Rep$  suggests a significant decrease in content production of users with an increase in reputation score after fair distribution (1 % increase in the  $Log(Rep)$  is expected to decrease by 0.045 Posts with HF 19. Our results also hold for analysis using the Negative Binomial, as presented in column (5) of Table 3.

Together, these analyses provide compelling evidence supporting Hypotheses 2a and 2b. The pattern reveals how the interplay between fairness in reward distribution, user resources, and content creation unfolds in the context of blockchain-based social media.

### 5.3. Checks for underlying mechanisms behind user engagement

Our key argument focuses on the dynamic relationship between user contributions, platform growth, and cryptocurrency rewards. On platforms like Steemit, user-generated, high-quality content attracts a broader audience, driving platform expansion. This not only enhances platform growth but also increases the value of the rewards received by users. In this subsection, we examine this mechanism, crucial to understanding how cryptocurrency rewards influence the growth of blockchain-based social media platforms. By exploring these connections, we aim to clarify the complexities of reward-driven user contributions.

#### 5.3.1. Do users care about the platform's growth?

Our data includes time-stamped user activities, allowing us to examine preferences for tradable and non-tradable cryptocurrencies on Steemit. Users can convert STEEM to non-tradable SP through an option called Power-Up, enhancing their voting power and influence over content rewards. This choice is seen as an investment in the platform rather than short-term profit-seeking. This analysis helps us understand the motivations behind user-generated content on crypto-integrated platforms, providing insights into whether users are more committed to the platform or driven by immediate financial gains.

We first generate a binary indicator and assign 1 if the user ever chose *Power-Up* option to convert STEEM currency to SP within the

user's account, which we call *PowerUp*. Otherwise, the variable takes a value of 0. We also consider an alternative variable, *PowerUpAmount*, that measures the user's total number of exchanges made within the user account during a given week. Afterward, we estimate Eq. (1) by replacing  $y_{it}$  with either *PowerUp* or *PowerUpAmount* variable to examine whether users are more likely to prefer to invest in the platform's growth over to seek for short-term profit.

The results of this analysis are summarized in columns (1)–(3) of Table 4. As depicted in column (2), the introduction of HF19 (reflected by the positive and statistically significant coefficient  $\beta = 1.003$ ,  $p < 0.01$ ) has a meaningful effect on users' inclination to choose the 'PowerUp' option. This effect remains consistent when we consider an alternative outcome variable, *PowerUpAmount*, as displayed in column (3) ( $\beta = 0.501$ ,  $p < 0.01$ ). Collectively, these findings suggest that the transition towards a fairer crypto-reward distribution made under HF19 led users to be more invested in the platform's growth. This represents a shift away from short-term profit-seeking behavior, indicating a greater commitment to contributing to the platform's community and success. This aligns with the underlying mechanism uniquely pertaining to crypto-integrated social media. This further corroborates the unique dynamics of user motivation and engagement on crypto-integrated social media platforms like Steemit.

### 5.3.2. Do users generate higher quality contents?

Commitment of users to the online community within a social media platform is associated with the quality of content created by them; e.g., Chen et al. (2011) report that users who are more engaged in the online community are more likely to contribute to the platform by creating high-quality posts [35]. Hence, we explore the impact of fairer reward distribution on the quality of user content.

On the Steemit platform, users can vote for content generated by others, with the number of votes typically serving as an indication of content quality. With this in mind, we use the total number of votes received by individual user  $i$  in a given week as a proxy for content quality [9], which we call this measure *UpVote*. Additionally, we explicitly extract post-specific features to measure *UpVote*. Additionally, we explicitly extract post-specific features to measure the quality based on previous literature: *AvgLength* [9,36], *TagCnt* [36], *LinkCnt* [37], and *ImgCnt* [38,39]. They are defined as the average length of text, the average number of tags, links, and images on posts produced by user  $i$  in the same period respectively. We then apply Eq. (1) by replacing  $y_{it}$  with content quality measures to assess whether users were more inclined to generate high-quality content after HF19.

Our findings are presented in Table 5. As indicated in column (2), HF19 is associated with a notable positive impact on content quality, with the coefficient  $\beta = 0.214$ ,  $p < 0.01$  being statistically significant. This effect remains stable when we substitute the dependent variable with the other content quality variables, as shown in columns (2)–(6). These results imply that the shift towards a fair distribution under HF19

**Table 4**

Effects of HF19 on non-tradable cryptocurrency use.

| Variables       | <i>PowerUp</i><br>(1) Logit | <i>PowerUpAmount</i><br>(2) OLS |
|-----------------|-----------------------------|---------------------------------|
| <i>Treat</i>    | 1.003***<br>(0.056)         | 0.501***<br>(0.032)             |
| <i>Log(SP)</i>  | 0.063***<br>(0.008)         | 0.019***<br>(0.003)             |
| <i>Log(Rep)</i> | 0.041***<br>(0.014)         | 0.008***<br>(0.003)             |
| Controls        | Yes                         | Yes                             |
| Fixed Effects   | Yes                         | Yes                             |
| Log-likelihood  | −16,971                     | –                               |
| R-squared       | –                           | 0.033                           |
| Observations    | 62,790                      | 62,916                          |

Notes. Standard errors in parentheses. \*\*\*  $p < 0.01$ , \*\*  $p < 0.05$ , \*  $p < 0.1$ .

**Table 5**

Effects of HF19 on content quality.

| Variables       | <i>Upvote</i><br>(1) Poisson | <i>AvgLength</i><br>(2) OLS | <i>TagCnt</i><br>(3)<br>Poisson | <i>LinkCnt</i><br>(4)<br>Poisson | <i>ImgCnt</i><br>(5)<br>Poisson |
|-----------------|------------------------------|-----------------------------|---------------------------------|----------------------------------|---------------------------------|
| <i>Treat</i>    | 0.214***<br>(0.042)          | 1.396***<br>(0.060)         | 0.054**<br>(0.026)              | 0.194**<br>(0.088)               | 0.090**<br>(0.037)              |
| <i>Log(SP)</i>  | 0.017<br>(0.013)             | 0.089***<br>(0.007)         | 0.065***<br>(0.007)             | 0.033**<br>(0.016)               | 0.070***<br>(0.011)             |
| <i>Log(Rep)</i> | 0.064***<br>(0.020)          | 0.047*<br>(0.014)           | 0.049***<br>(0.015)             | 0.009<br>(0.028)                 | 0.060***<br>(0.019)             |
| Controls        | Yes                          | Yes                         | Yes                             | Yes                              | Yes                             |
| Fixed Effects   | Yes                          | Yes                         | Yes                             | Yes                              | Yes                             |
| Log-likelihood  | −2,335,662                   | –                           | −135,750                        | −87,943                          | −113,098                        |
| R-squared       | –                            | 0.058                       | –                               | –                                | –                               |
| Observations    | 62,916                       | 29,360                      | 62,874                          | 56,280                           | 61,362                          |

Notes. Standard errors in parentheses. \*\*\*  $p < 0.01$ , \*\*  $p < 0.05$ , \*  $p < 0.1$ .

enhanced users' likelihood of generating high-quality content. This insight aligns with the theoretical framework discussed in Section 2, further validating the unique dynamics that drive user motivation and engagement on crypto-integrated social media platforms like Steemit.

### 5.3.3. Do users with high resources also contribute more after HF19?

Supported by the comprehensive theoretical grounding offered by both the Fairness and Self-Determination Theories, it becomes imperative to explore how these mechanisms operate amidst a diverse user spectrum, particularly when categorizing users based on their financial and reputational resources [40]. In this section, we conduct a subsample analysis segmented by different user groups, illustrating that the observed effect remains consistent across the different groups. The following justifications underscore this investigation.

First, dissecting the effect across various financial segments ensures the behaviors documented aren't exclusively propelled by the affluent. Should our findings predominantly echo the sentiments of Minnows, the platform's predominant demographic who likely anticipated heightened financial returns post-HF19, it might insinuate that the impetus for content creation leans heavily towards capital gains rather than the intrinsic motivations fostered by a just rewards system. Second, evaluating the central effect across a spectrum of reputational metrics allows us to discern that both newcomers and seasoned participants are equivalently swayed by the new reward distribution paradigm. A lopsided effect skewed towards high-reputation users might insinuate the platform's ecosystem where entrenched members dominate, a scenario that could inhibit the platform's organic growth and vibrancy.

Our findings in Table 6 show that the transition towards a fair reward distribution, as instated under HF19, leads to more content creation, even among users with abundant financial and reputational resources. Such consistency across demographics further validates that the psychological and motivational nuances introduced by transparent crypto-reward distribution stand as important foundational factors influencing user behavior and platform engagement.

We further conduct a detailed subsample analysis focusing on SP acquired by users through their contributions to the Steemit platform. SP can be obtained in two primary ways. The first is through contribution-based activities, where users earn SP as a reward for creating and curating content, thereby accumulating influence through their participation. The second method involves purchasing SP by converting STEEM to SP, as illustrated in Fig. 1. This distinction in SP acquisition methods allows for a more precise examination of the effects of reward distribution changes on different user subgroups based on their SP source. For this analysis, we define contribution-based SP using the following formula:

$$ContributionSP_{it} = \sum_{t=1}^{t-1} AuthorReward_{i,t-1} + \sum_{t=1}^{t-1} CurationReward_{i,t-1}. \quad (3)$$

**Table 6**

Subsample Analysis: Steem Power and Reputation.

| Groups         | SP 75Q              |                     | SP 90Q              |                     | Rep 75Q             |                     | Rep 90Q             |                     |
|----------------|---------------------|---------------------|---------------------|---------------------|---------------------|---------------------|---------------------|---------------------|
| Variables      | (1) Post            | (2) Length          | (3) Post            | (4) Length          | (5) Post            | (6) Length          | (7) Post            | (8) Length          |
| Treat          | 0.447***<br>(0.041) | 1.638***<br>(0.126) | 0.551***<br>(0.089) | 1.809***<br>(0.206) | 0.443***<br>(0.379) | 1.603***<br>(0.121) | 0.336***<br>(0.042) | 1.224***<br>(0.156) |
| Controls       | Yes                 | Yes                 | Yes                 | Yes                 | Yes                 | Yes                 | Yes                 | Yes                 |
| Fixed Effects  | Yes                 | Yes                 | Yes                 | Yes                 | Yes                 | Yes                 | Yes                 | Yes                 |
| Log-likelihood | −30,478             | —                   | −11,654             | —                   | −38,881             | —                   | −16,772             | —                   |
| R-squared      | —                   | 0.029               | —                   | 0.018               | —                   | 0.077               | —                   | 0.097               |
| Observations   | 15,729              | 6300                | 15,729              | 6300                | 15,729              | 6300                | 15,729              | 6300                |

Notes. Standard errors in parentheses. \*\*\*  $p < 0.01$ , \*\*  $p < 0.05$ , \*  $p < 0.1$ .

We then redo the analysis with the subsample by SP group and show that the results remain qualitatively the same, as shown in Table 7.

## 6. Robustness studies and alternative explanations

### 6.1. Treatment exogeneity

Note that we construct our treatment based on the date when HF19 was implemented. One potential concern could be that this treatment is not exogenous, given that HF19 represents a deliberate modification of the platform's rules and functionality. This change was designed to address existing issues, enhance performance, and align with the platform's overarching goal of equality. However, it is important to note that a hardfork in the context of the Steemit platform, such as HF19, represents an external shift not influenced by the internal mechanisms of the platform itself. The decision to implement a hardfork emerges outside the routine operation of the platform, typically made by the developers or the governing body associated with Steemit. Steem operates on a Proof-of-Stake (PoS) system, where major decision-making processes are handled by a total of 21 witnesses, a user group elected by votes, providing further support to our argument of exogeneity [41].

We base our treatment on HF19's implementation date, considered exogenous despite being a deliberate rule modification aimed at enhancing platform equality. HF19, as a Steemit hardfork, is an external change decided by developers or the governing body, not influenced by internal platform mechanisms. Steem's governance involves 21 elected witnesses, affirming the exogeneity of our approach [41]. Further, our tests using previous user activity as predictors show no significant correlations with HF19 in Table 8, confirming that changes in content creation i.e., (i.e.,  $\Delta Post$ ) are not predictively linked to the hardfork implementation.

### 6.2. Falsification test

We further examine if our treatment, the implementation of HF19, truly leads to an increase in content creation or if there is simply an upward trend over time. Our incorporation of crypto-related control variables (such as *STMPPrice*) largely helps account for such time trends. Fig. 2 illustrates that the pronounced increase in content production

**Table 7**

Subsample analysis: contributed steem power.

| Groups         | Contribution SP 75Q |                     | Contribution SP 90Q |                   |
|----------------|---------------------|---------------------|---------------------|-------------------|
| Variables      | (1) Post            | (2) Length          | (3) Post            | (4) Length        |
| Treat          | 0.248***<br>(0.051) | 0.275***<br>(0.080) | 0.222***<br>(0.067) | 0.222*<br>(0.126) |
| Controls       | Yes                 | Yes                 | Yes                 | Yes               |
| Fixed Effects  | Yes                 | Yes                 | Yes                 | Yes               |
| Log-likelihood | −30,680             | —                   | −12,375             | —                 |
| R-squared      | —                   | 0.080               | —                   | 0.069             |
| Observations   | 15,729              | 9533                | 6300                | 3889              |

Notes. Standard errors in parentheses. \*\*\*  $p < 0.01$ , \*\*  $p < 0.05$ , \*  $p < 0.1$ .**Table 8**

Logit regression to test of treatment exogeneity.

|                     | (1)              | (2)                | (3)               | (4)               |
|---------------------|------------------|--------------------|-------------------|-------------------|
|                     | 52 Weeks         | 52 Weeks           | 179 Weeks         | 179 Weeks         |
| Variables           | (2017 Year)      | (2017 Year)        | (2016–2019)       | (2016–2019)       |
| $\Delta Post$       | 0.019<br>(0.521) | −0.317<br>(0.697)  | −0.138<br>(0.192) | −0.118<br>(0.203) |
| $\Delta Upvote$     | 0.008<br>(0.016) | 0.001<br>(0.017)   | −0.000<br>(0.006) | −0.000<br>(0.006) |
| $\Delta SP$         | 1.461<br>(1.159) | 1.887<br>(1.274)   | −0.119<br>(0.164) | −0.123<br>(0.165) |
| $\Delta Rep$        | 0.661<br>(2.541) | −0.127<br>(2.690)  | −0.342<br>(0.473) | −0.343<br>(0.474) |
| $\Delta Following$  |                  | 0.005<br>(0.132)   |                   | −0.000<br>(0.052) |
| $\Delta Followed$   |                  | 0.315<br>(0.279)   |                   | −0.012<br>(0.062) |
| $\Delta Delegation$ |                  | 0.117<br>(0.124)   |                   | −0.001<br>(0.023) |
| $\Delta Conversion$ |                  | −1.236<br>(3.177)  |                   | −0.612<br>(2.354) |
| $\Delta Withdrawal$ |                  | 5.758<br>(4.433)   |                   | −0.179<br>(1.264) |
| $\Delta Deposit$    |                  | −12.205<br>(9.629) |                   | −0.317<br>(2.564) |
| Log-likelihood      | −34.41           | −32.63             | −113.16           | −113.33           |
| Observations        | 52               | 52                 | 179               | 179               |

Notes. Columns (1)–(3) is the results of Logit regression to predict treatment within the 2017 year. We then enlarge our sample to 179 Weeks spanning the 2016–2019 year in columns (3)–(5). Standard errors in parentheses. \*\*\*  $p < 0.01$ , \*\*  $p < 0.05$ , \*  $p < 0.1$ .

does not occur until after the treatment date. In addition, for further validation, we conduct a series of placebo tests. These tests redefine the treatment time based on the implementation of another policy, *Hardfork18*, in the 15th week of 2017, and a *similar upward trend* observed in the 5th week of 2018 (see Fig. 3). Since these treatments are false, we do not expect a positive and significant impact on user engagement.

The results (Table 9) show that the effects of false treatment are insignificant or non-positive. In columns (1)–(4), the coefficient of the false treatment variable is indeed insignificant, which aligns with the expectation that there should be no similar positive impact of the false treatment. Similarly, in columns (4)–(7), we observe a significant and negative coefficient for the false treatment, suggesting that content creation activities actually decreased during the period associated with the false treatment. Overall, these results reinforce our main findings by demonstrating the robustness of our model and ensuring that our conclusions are not influenced by unrelated temporal factors.

### 6.3. Alternative time windows

Our main sample includes active users during the ten weeks leading up to and following the implementation of HF19. We chose this range to focus on the most immediate effects of HF19 and to observe the



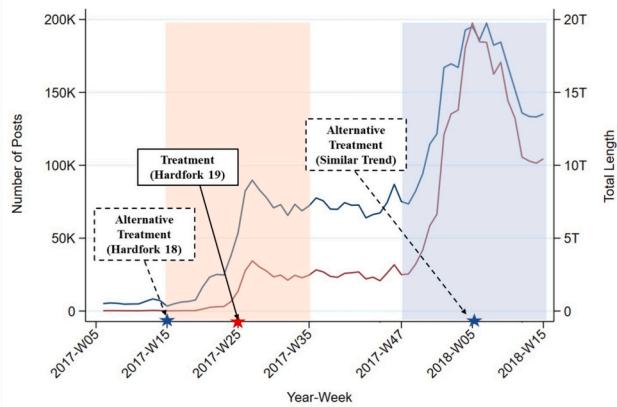


Fig. 3. Alternative (Placebo) treatments.

Note: Number of Posts (Blue); Total Length (Red). Alternative treatment dates are the 15th week, 2017 (Hardfork18), and the fifth week, 2018, which has the homogeneous trend in our dependent variables (preexisting upward and downward and stagnant trend).

adjustments in user behavior in response to the new reward system. However, this strategy may limit our perspective and could potentially overlook longer-term effects or trends that might develop outside this time frame. Furthermore, there may be other unobserved events or factors that could influence user behavior within this narrow window of time, making our findings specific to the sampled period and potentially confounding the effects of HF19.

To address these potential concerns, we also examined user activity in two other sample periods. The first alternative sample captures a shorter time span, including only the active users five weeks before and after HF19. The second adjusted sample, on the other hand, broadens the period to include users active 20 weeks before and 20 weeks after the implementation of HF19. Overall, as shown in Table 10, regardless of the sample period considered, our results remained consistent with our main conclusions, affirming the robustness of our findings.

#### 6.4. Other alternative explanations

In this section, we address other alternative explanations to ensure the robustness of our findings.<sup>2</sup> One other concern is the influence of broader trends in the cryptocurrency market on our results. For example, significant fluctuations in the value of a leading cryptocurrency like Bitcoin could impact user engagement on crypto-integrated platforms such as Steemit [42]. To mitigate this, we have incorporated additional control variables related to Bitcoin, capturing the overall market dynamics. We confirm that our findings are not simply reflections of broader market movements but are indeed indicative of the specific impacts of policy changes within the Steemit platform.

Another important aspect of our study is the consideration of the various changes introduced by Hardfork19 (HF19), each with potential implications for user behavior and platform dynamics on Steemit. Notably, HF19 included significant modifications to the voting impact and curation rewards. The voting impact was substantially increased, making each vote more powerful but less frequent. However, our analysis of post-HF19 data showed an increase in upvotes, suggesting that these changes in voting dynamics did not adversely affect user engagement. Additionally, we considered the possibility of new users being attracted by the modified curation rewards. We limited our analysis to users active on Steemit before HF19, ensuring that our findings reflected

the behavioral changes of existing users, rather than the influence of new users adapting to the updated platform dynamics.

Another possible concern is that the observed increase in posting activity could indeed be attributed to the growth in the value of rewards through increased Steem prices. To address this concern, we conduct additional analysis, considering a different sample period characterized by a consistent downward trend in coin prices (from 23rd to 28th week, 2017) and replicate our analysis for this period. If our results were driven by the coin price trend, we would not expect the positive impact of HF19. However, the results are consistent with our main findings, reinforcing our conclusion that the observed increase in posting activity cannot be solely attributed to changes in monetary incentives.

## 7. Conclusion and discussion

To our knowledge, this paper represents the first empirical investigation into the effect of fairness in crypto-based incentives on user-generated content in a social media environment. Utilizing a system update on the Steemit platform (i.e., HF19), we scrutinize the behavioral changes of users in their voluntary posting activities, a key contribution to the platform's growth and vitality. Our findings indicate that the transition to a fairer rewards distribution system results in a substantial increase in the volume of posts, with this increase more pronounced among users with lower capital and reputational resources. These users, previously holding less sway in the rewards distribution, are more motivated to contribute content under the more equitable system. Conversely, high-resource users, who traditionally commanded a larger share of the rewards based on their financial and influential clout, may perceive this fair distribution as relatively unjust, given their significant pre-existing resources and status. Nevertheless, we observe the increased trend of content creation across the different user groups by financial and reputational resources.

### 7.1. Theoretical contributions

Our paper makes meaningful contributions to the literature on financial incentives in online platforms. Prior research has largely concentrated on traditional financial incentives [9,13,14,34,43,44]. In contrast, our study furnishes a nuanced understanding of crypto-based incentives, emphasizing the interconnectedness between user contributions and platform growth. Moreover, our exploration of rewards allocation within online platforms—through the lens of a platform-wide (top-down) distributive rule—significantly advances this domain of research. While previous studies have examined rewards allocation, such as the distribution between responders and proposers in online referral systems [27], and the use of probabilistic, uncertain rewards in crowdfunding platforms [45], our research underscores the importance of reward distribution mechanisms in online social media. We demonstrate their impact on overall user engagement and platform growth, shedding light on the effectiveness of an innovative financial incentive—the crypto-based incentive system.

Our study also enriches the field of distributive justice. While distributive justice is a fundamental theory in organizational fields, playing a pivotal role in motivating members and enhancing performance [3], existing research has been predominantly behavior-oriented [24,46,47], bound by the lack of empirical rigor. We highlight the significance of leveraging the unique advantages of crypto-based platforms—observable rewards and algorithmic distributive rules—to empirically illuminate the effects of achieving distributive justice in social media.

### 7.2. Practical implications

Our study has significant implications for stakeholders in crypto-integrated social media platforms. Many platforms use hierarchical reward systems that initially attract investment and high-quality

<sup>2</sup> Due to page limit, full reports are not included but are available upon request.

**Table 9**

Falsification test using alternative treatment dates.

| Variables         | Hardfork18        |                   |                   | Similar Upward Trend |                      |                     |
|-------------------|-------------------|-------------------|-------------------|----------------------|----------------------|---------------------|
|                   | (15th week, 2017) |                   |                   | (5th Week, 2018)     |                      |                     |
|                   | Post              | Length            |                   | Post                 | Length               |                     |
|                   | (1) Poisson       | (2) NB            | (3) OLS           | (4) Poisson          | (5) NB               | (6) OLS             |
| Alternative Treat | −0.004<br>(0.030) | −0.012<br>(0.023) | −0.058<br>(0.042) | −0.196***<br>(0.020) | −0.164***<br>(0.004) | −0.444**<br>(0.029) |
| Controls          | Yes               | Yes               | Yes               | Yes                  | Yes                  | Yes                 |
| Fixed Effects     | Yes               | Yes               | Yes               | Yes                  | Yes                  | Yes                 |
| Log-likelihood    | −66,884           | −55,396           | −                 | −807,355             | −688,150             | −                   |
| R-squared         | —                 | —                 | 0.170             | —                    | —                    | 0.012               |
| Observations      | 51,198            | 51,198            | 24,334            | 409,552              | 409,552              | 266,842             |

Notes. Standard errors in parentheses. \*\*\*  $p < 0.01$ , \*\*  $p < 0.05$ , \*  $p < 0.1$ . As with the main analysis, we set the sample period to 21 weeks, with 10 weeks before and after the treatment. For the Hardfork 18 sample, this period spans from the 5th week, of 2017 to the 25th week, of 2017. The similar upward trend sample period spans from the 47th, of 2017 to the 15th week, of 2018.

**Table 10**

Sensitivity test using various time windows.

| Alternative Window | 11 Weeks (20th – 30th Week, 2017) |                     |                     | 41 Weeks (5th – 45th Week, 2017) |                     |                     |
|--------------------|-----------------------------------|---------------------|---------------------|----------------------------------|---------------------|---------------------|
|                    | Post                              |                     |                     | Post                             |                     |                     |
|                    | (1) Poisson                       | (2) NB              | (3)                 | (4) Poisson                      | (5) NB              | (6)                 |
| Treat              | 0.175***<br>(0.039)               | 0.248***<br>(0.020) | 0.473***<br>(0.114) | 0.504***<br>(0.033)              | 0.371***<br>(0.016) | 1.210***<br>(0.059) |
| Controls           | Yes                               | Yes                 | Yes                 | Yes                              | Yes                 | Yes                 |
| User Fixed         | Yes                               | Yes                 | Yes                 | Yes                              | Yes                 | Yes                 |
| Log-likelihood     | −68,082                           | −55,591             | −                   | −177,034                         | −138,015            | −                   |
| R-squared          | —                                 | —                   | 0.038               | —                                | —                   | 0.133               |
| Observations       | 36,619                            | 36,619              | 17,497              | 118,162                          | 118,162             | 68,163              |

Notes. Standard errors in parentheses. \*\*\*  $p < 0.01$ , \*\*  $p < 0.05$ , \*  $p < 0.1$ .

contributors by skewing rewards towards users with greater stake and influence. However, this approach can lead to a perceived unfairness as platforms grow, especially when reward disparities become visible, as in crypto platforms. Our analysis post-HF19 shows that rewards became more equitable, reducing the concentration of rewards among the top 5 % of posts from 76 % to 60 %, which can foster broader user engagement and platform growth.

Despite these positive changes, resistance from high-resource users, or ‘Whales,’ is to be expected. To manage this, we suggest a hybrid reward function that starts with a rapid ascent and transitions to a linear function beyond a certain threshold. This design encourages substantial initial rewards but prevents excessive wealth concentration, promoting fairness across the user base.

Our findings offer valuable insights for designing incentive mechanisms on crypto-based social media platforms, providing a foundation for sophisticated, data-driven tools that predict user responses and guide equitable incentive policies. By integrating distributive justice into platform algorithms, our work supports enhanced user engagement and platform sustainability. This approach is essential for evolving decision-making processes in the digital realm, ensuring they remain effective and responsive.

### 7.3. Limitation and future work

While this study provides valuable insights into fair reward distribution in crypto-integrated social media platforms, focusing only on Steemit may limit generalizability. Different platforms may have unique reward systems or user bases responding differently to reward changes. We analyze the impact of HF19, a major change in Steemit's reward distribution, to study fairness. However, reward systems are complex, and fairness needs more comprehensive future analysis beyond a single policy. Moreover, our fairness metric is based on distribution equality, but other aspects like merit or effort could also be crucial. We measure user engagement by textual content, potentially missing impacts from

non-textual content like videos and images. Future research should explore these dimensions to better understand user engagement.

### CRedit authorship contribution statement

**Woojin Yang:** Writing – original draft, Visualization, Formal analysis, Data curation, Conceptualization. **Yeongin Kim:** Writing – review & editing, Supervision, Methodology, Conceptualization. **Taehun Kim:** Validation, Resources, Investigation. **Chul Ho Lee:** Writing – review & editing, Validation, Supervision, Project administration, Conceptualization. **Yasin Ceran:** Writing – review & editing, Visualization, Methodology.

### Declaration of competing interest

None.

### Data availability

Data will be made available on request.

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**Woojin Yang** is a Ph.D. candidate at KAIST, and his interest includes the social impact of platform business and blockchain tokenization. He also extends his area to the combinatorial studies of national language processing and econometrics.

**Yeongin Kim** is an Assistant Professor at the Wake Forest School of Business at Wake Forest University. Before joining Wake Forest, he worked as an Assistant Professor at Virginia Commonwealth University. He earned his M.S. from Texas A&M University and his Ph.D. from the University of Texas at Dallas. His research primarily focuses on the economics of information technology (IT) in various settings. Specifically, he investigates the value of information sharing in decision conflicts, the risks posed by widespread IT, strategies to reduce these risks, and the use of Artificial Intelligence (AI) to enhance operational efficiency. To explore these subjects, he employs game theory, econometrics, and optimization models. His recent research has been featured in *MIS Quarterly*, *Information Systems Research*, and *Decision Support Systems*.

**Tae Hun Kim** is an Associate Professor at Incheon National University, South Korea. He received his M.S. from Korea University and his Ph.D. from Michigan State University. His research centers on data science, business analytics, and the value of technology. Recently, his work has been published in prominent Information Systems journals, including *Information Systems Research*, *European Journal of Information Systems*, and *Information & Management*, among others.

**Chul Ho Lee** is an Associate Professor at the School of Business and Technology Management at KAIST. He obtained his Ph.D. in Management Science at the University of Texas at Dallas. He worked as a faculty member at the Harbin Institute of Technology in China and Xavier University in the US. He published his papers in top journals such as *Information Systems Research* and *Production and Operations Management* as well as ABDC A\* journals, including *Computers in Human Behavior* and *Information & Management*. His research covers the economics of emerging technologies.

**Yasin Ceran** is an associate professor at KAIST University, Korea. He completed his Ph.D. in Management Information Systems from the University of Texas at Dallas. Yasin has worked vigorously on an array of data-related projects encompassing data mining, statistics, and modeling and is dedicated to sharing his experience and expertise with learners. His works have been published in *Management Science*, *Production and Operations Management*, and *Computers in Human Behavior*.